

A Product Dichotomy for Separable C^* -Algebras and BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. It asks for a noncommutative C^* -algebra version of the commutative $C(K)$ BCP criterion:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1. *Let $(A_k)_{k \geq 1}$ be separable C^* -algebras. Then*

$$A = \prod_{k=1}^{\infty} A_k$$

with the C^ -product norm has UBCP.*

Theorem 2. *Let $(A_\gamma)_{\gamma \in \Gamma}$ be a family of C^* -algebras with uncountably many nonzero factors. Then*

$$A = \prod_{\gamma \in \Gamma} A_\gamma$$

has countable capture. Consequently A fails BCP.

Corollary 1. *Let $(A_\gamma)_{\gamma \in \Gamma}$ be a family of separable C^* -algebras. Then*

$$\prod_{\gamma \in \Gamma} A_\gamma$$

has BCP, equivalently UBCP, if and only if at most countably many factors are nonzero.

Corollary 2. *There are nonseparable non-Type-I C^* -algebras with UBCP. For example,*

$$\ell_\infty(\mathcal{O}_2) = \prod_{k=1}^{\infty} \mathcal{O}_2$$

has UBCP.

Idea of the Proof

The product is large, but a unit vector in a sup norm product must be large in some coordinate. In that coordinate we use separability to choose a countable net of sphere directions. A centre supported in one coordinate has norm exactly 2. All other coordinates contribute distance at most 1, and the chosen coordinate contributes distance less than $3/2$. Thus the unit sphere is covered by countably many balls of radius $3/2$, each centred at distance 2 from the origin.

The negative direction is also coordinate-theoretic. Given countably many old centres in an uncountable product, each centre has at most one coordinate whose deletion lowers its norm. Avoid the countable set of such coordinates, put a unit vector in the remaining coordinate, and project away from that coordinate. The projection kills the new vector but preserves the norms of all old centres, which is exactly the countable-capture obstruction.

Proof

Proof of Theorem 1. If $A_k = 0$, ignore that coordinate. For each nonzero A_k , choose a countable $1/16$ -net D_k in the unit sphere of A_k .

For each pair (k, d) , with $d \in D_k$, define a product element

$$c_{k,d} = (0, \dots, 0, 2d, 0, \dots),$$

supported only in coordinate k . Then

$$\|c_{k,d}\| = 2.$$

We claim that the balls

$$B(c_{k,d}, 3/2)$$

cover S_A . Let $x = (x_j) \in S_A$. Choose k such that

$$\|x_k\| > 3/4.$$

Pick $d \in D_k$ with

$$\left\| d - \frac{x_k}{\|x_k\|} \right\| < \frac{1}{16}.$$

For coordinates $j \neq k$,

$$\|x_j - c_{k,d,j}\| = \|x_j\| \leq 1 < 3/2.$$

In coordinate k ,

$$\begin{aligned}\|x_k - 2d\| &\leq \left\| x_k - 2 \frac{x_k}{\|x_k\|} \right\| + 2 \left\| \frac{x_k}{\|x_k\|} - d \right\| \\ &= 2 - \|x_k\| + \frac{1}{8} \\ &< 2 - \frac{3}{4} + \frac{1}{8} = \frac{11}{8} < \frac{3}{2}.\end{aligned}$$

Therefore $\|x - c_{k,d}\| < 3/2$. The family of pairs (k, d) is countable, so these countably many balls cover S_A . Since every centre has norm 2 and every radius is $3/2$, the balls are uniformly separated from the origin. Hence A has UBCP. $\square$

Proof of Theorem 2. Let $C \subset A$ be countable. For $c = (c_\gamma)_{\gamma \in \Gamma} \in A$, call a coordinate γ essential for c if

$$\sup_{\eta \neq \gamma} \|c_\eta\| < \|c\|.$$

For fixed c , there is at most one essential coordinate: if γ is essential, then necessarily $\|c_\gamma\| = \|c\|$, and no second coordinate can also have norm $\|c\|$.

The union of the essential-coordinate sets for $c \in C$ is therefore countable. Since uncountably many factors are nonzero, choose a nonzero coordinate γ_0 outside this union. Pick $v \in A_{\gamma_0}$ with $\|v\| = 1$, and let $u \in A$ be supported only at γ_0 , with $u_{\gamma_0} = v$.

Let $P : A \rightarrow \prod_{\gamma \neq \gamma_0} A_\gamma$ be the coordinate-deletion map. Then P is contractive and $P(u) = 0$. Because γ_0 is not essential for any $c \in C$, we have $\|P(c)\| = \|c\|$. Hence, for every $\lambda \in \mathbb{C}$,

$$\|c - \lambda u\| \geq \|P(c - \lambda u)\| = \|P(c)\| = \|c\|.$$

Taking the infimum over λ gives $\text{dist}(c, \mathbb{C}u) \geq \|c\|$, while equality follows from $\lambda = 0$. Thus A has countable capture by the countable-capture characterization, and the countable-capture obstruction implies failure of BCP. $\square$

Proof of the product dichotomy. If at most countably many factors are nonzero, the product reduces to a countable product of separable C^* -algebras and has UBCP by Theorem 1. If uncountably many factors are nonzero, then Theorem 2 gives countable capture, hence failure of BCP. Since UBCP implies BCP, the equivalence follows. $\square$

Scope

This is a positive counterpart to the uncountable tensor-product and corona obstructions, but it also gives the exact answer for products of separable C^* -algebras. It shows that nonseparability and non-Type-I behaviour alone cannot force BCP failure: $\ell_\infty(\mathcal{O}_2)$ is positive. What matters for products is whether the index set of nonzero factors is countable or uncountable.

Novelty and Search Bounds

The local indexes were searched for countable products, uncountable products, ℓ_∞ -sums, separable C^* -algebras, BCP, and UBCP. Web searches on June 17, 2026 for countable products and ball-covering property found nearby stability literature but no exact C^* -product dichotomy packet. The proof here is self-contained.

Human Review Focus

Reviewers should check:

1. the coordinate choice $\|x_k\| > 3/4$;
2. the $11/8 < 3/2$ estimate;
3. the uniform off-origin constants $\|c_{k,d}\| = 2$ and radius $3/2$;
4. the assertion that each old product-centre has at most one essential coordinate whose deletion lowers its norm.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.